# Flow Matching and Continuous-Time Generative Models

**Flow Matching (FM)** is a simulation-free framework for training Continuous Normalizing Flows (CNFs) that avoids the costly ODE simulation required by maximum-likel training. Instead of back-propagating through an ODE solver, FM directly regresses a neural network $v_\theta(x, t)$ against analytically tractable *conditional* vector fields $u_t($

FM subsumes diffusion models as a special case — the standard Gaussian diffusion path is recovered by choosing $\alpha_t = 1 - t, \sigma_t = t$. More importantly, FM enables **Opt Transport (OT) paths** that move particles along straight lines ($x_t = (1 - t)x_0 + tx_1$), yielding straighter probability-flow ODEs that converge in far fewer function eval ≈ 142 vs ≈ 1 000 for DDPM) while matching or improving NLL and FID on standard image benchmarks.

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## Background: Generative Modeling in Continuous Time

### Continuous Normalizing Flows (CNFs)

A CNF transforms a simple base distribution $p_0$ into a complex data distribution $p_1$ by integrating a time-dependent vector field $v_t : \mathbb{R}^d \times [0, 1] \to \mathbb{R}^d$:

$$\frac{d}{dt} f_t(x) = v_t\big(f_t(x),\, t\big), \qquad f_0(x) = x$$

The induced density $p_t$ satisfies the **continuity equation**:

$$\frac{\partial p_t}{\partial t} = -\nabla \cdot (p_t\, v_t)$$

**Maximum-likelihood training** of CNFs requires:

1. Integrating the ODE forward to obtain $f_1(x)$,
2. Estimating $\log p_1$ via the instantaneous change-of-variables formula $\frac{d}{dt} \log p_t(f_t(x)) = -\nabla \cdot v_t(f_t(x), t)$,
3. Back-propagating *through the ODE solver* — expensive and numerically unstable.

This quadratic-to-cubic cost in the number of ODE steps made direct MLE training of large CNFs largely impractical before Flow Matching.

### Score-Based Diffusion Models

Diffusion models define a **forward noising process**:

$$z_t = \alpha_t\, x_1 + \sigma_t\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I)$$

where $\alpha_t \searrow 0$ and $\sigma_t \nearrow 1$ as $t : 0 \to 1$. A neural network learns the **score** $\nabla_{z_t} \log p_t(z_t)$ via the denoising loss:

$$\mathcal{L}_{\mathrm{DSM}}(\theta) = \mathbb{E}_{t,\, x_1,\, \varepsilon}\left[\left\| s_\theta(z_t, t) + \frac{\varepsilon}{\sigma_t} \right\|^2\right]$$

Sampling proceeds via the **reverse SDE** (DDPM, ≈ 1 000 steps) or the corresponding **probability-flow ODE** (DDIM), which requires 250–1 000 network evaluations for hi samples. Each evaluation is a full forward pass of a large U-Net, making diffusion inference slow in practice.

### Diffusion as a Special Case of Flow Matching

With the **Gaussian path** $\alpha_t = 1 - t, \sigma_t = t$ the marginal $p_t(x) = \int \mathcal{N}(x;\, (1-t)x_1,\, t^2 I)\, q_1(x_1)\, dx_1$ recovers exactly the forward diffusion marginals.

The corresponding **probability-flow ODE** vector field is:

$$u_t(x) = \frac{\dot{\alpha}*t*}{\alpha_t}*(x - \sigma_t* s\theta(x,t))

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+ \dot{\sigma}_t \, s_\theta(x,t)$$
```

DDIM is precisely **Euler integration** of this ODE — so diffusion sampling *is* flow-matching sampling under the Gaussian path. Flow Matching generalises this by allowinc family, enabling the straight OT paths that dramatically reduce NFE.

# Flow Matching: Formulation and Objective

## The Flow Matching Loss

Given a **probability path** $\{p_t\}_{t\in[0,1]}$ with $p_0 = \mathcal{N}(0, I)$ and $p_1 \approx q_{\text{data}}$, let $u_t(x)$ be the **marginal vector field** that generates $p_t$. Flow Matching trains $v_\theta$ by minimising:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}[0,1],\, x\sim p_t}\left[\|v_\theta(x,t) - u_t(x)\|^2\right]$$

**Problem**: computing $u_t(x) = \mathbb{E}_{x_1\sim p_{1|t}(x_1|x)}[u_t(x|x_1)]$ requires the intractable posterior $p_{1|t}$.

The key insight of *Conditional* Flow Matching (CFM) is to bypass this by regressing on the analytically known **conditional** fields $u_t(x \mid x_1)$ instead.

## Probability Path Families

**Gaussian diffusion path** (Lipman et al. 2022, Section 2):

$$p_t(x \mid x_1) = \mathcal{N}\big(x;\, \mu_t(x_1),\, \sigma_t^2 I\big)$$

Trajectories are **curved** because noise is mixed in at all scales simultaneously.

**Optimal-Transport path** (same paper, Section 3):

$$x_t = (1-t)\,x_0 + t\,x_1, \qquad x_0 \sim \mathcal{N}(0, I)$$

The conditional vector field is **constant along each trajectory**:

$$u_t(x \mid x_1) = x_1 - x_0$$

OT paths are **straight lines** in sample space — the ODE trajectory has zero curvature, so very few Euler steps suffice for accurate simulation.

## Conditional Flow Matching (CFM) — Theorem 2

Define the **conditional path** $p_t(x \mid x_1) = \mathcal{N}(x;\, \mu_t(x_1), \sigma_t^2 I)$. The analytic conditional velocity is:

$$u_t(x \mid x_1) = \dot{\mu}_t(x_1) + \frac{\dot{\sigma}_t}{\sigma_t}\big(x - \mu_t(x_1)\big)$$

The **Conditional Flow Matching loss** replaces $u_t(x)$ with $u_t(x \mid x_1)$:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,\, x_1\sim q_1,\, x\sim p_t(\cdot|x_1)}\left[\|v_\theta(x,t) - u_t(x \mid x_1)\|^2\right]$$

**Theorem 2** (Lipman et al. 2022): The two losses share identical gradients,

$$\nabla_\theta\, \mathcal{L}_{\text{CFM}}(\theta) = \nabla_\theta\, \mathcal{L}_{\text{FM}}(\theta)$$

because the difference $\|u_t(x) - u_t(x \mid x_1)\|^2$ is independent of $\theta$. This means we can train with the tractable CFM objective and recover the optimal marginal vector fi
**simulation required during training**.

# Interactive Visualization: Flow Paths

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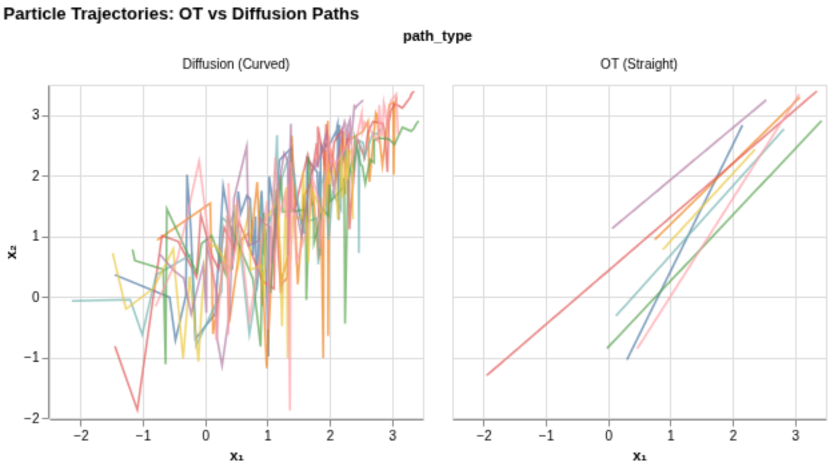
made with marimo

Particle Trajectories: OT vs Diffusion Paths




*OT paths move particles in straight lines (left), while diffusion paths create curved trajectories (right). Straight paths require fewer ODE solver steps.*

## Comparison to Score/Diffusion Models

### Training Loss Equivalence

Under the **Gaussian path** ($\mu_t(x_1) = (1-t)x_1, \sigma_t = t$) the CFM loss reduces to exactly the **denoising score-matching** objective used by DDPM/EDM. Both methods regr optimal vector field; FM simply exposes this without the need to interpret the target as a score.

### Sampling Speed

| Method | Path | Typical NFE |
|---|---|---|
| DDPM | Gaussian | 1 000 |
| DDIM | Gaussian (ODE) | 250 |
| FM – Gaussian | Gaussian | 270 |
| FM – OT | Straight line | **142** |

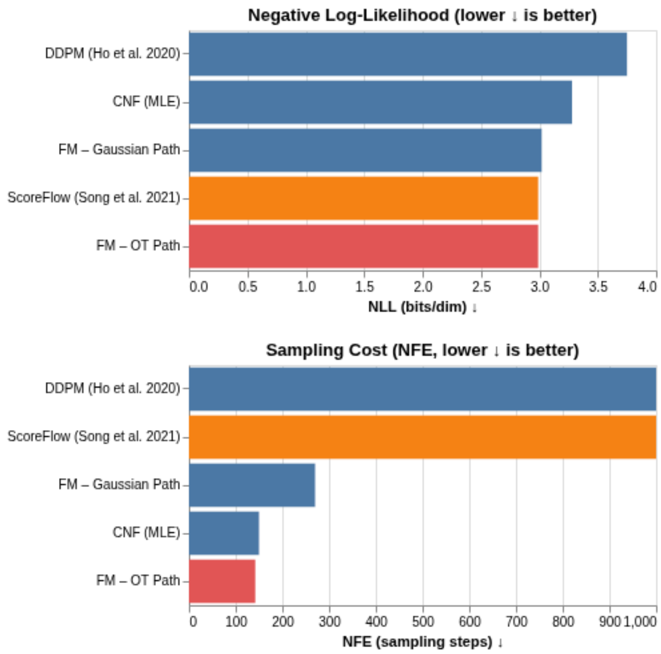
Because OT trajectories are straight, the probability-flow ODE has low curvature and a simple Euler or RK4 solver converges with far fewer steps.

### Training Efficiency

Lipman et al. (2022) train FM on ImageNet-128 in **500 K iterations** vs **4.36 M** for an equivalent diffusion baseline — an 8× reduction — with better or matching NLL. The free objective eliminates the per-step ODE solve that dominates diffusion training wall-clock time.

```
/tmp/marimo_2978/__marimo__cell_qnkX_.py:29: UserWarning: Automatically deduplicated selection parameter with identical configuration. If you want
  comparison_charts = (nll_chart & nfe_chart).resolve_scale(color="shared")
```

### Graph 1: Model Comparison on CIFAR-10




*Data from Lipman et al. (2022). FM-OT achieves lowest NFE (142 vs 1000 for DDPM) with comparable NLL.*

| Model<br>str | Path<br>str | NLL (BPD)<br>float64 | FID<br>float64 | NFE<br>int64 | Training Iters<br>str | Stability<br>str |
|---|---|---|---|---|---|---|
| DDPM (Ho et al. 2020) | Gaussian | 3.75 | 3.17 | 1,000 | 800K | Good |
| ScoreFlow (Song et al. 2021) | Gaussian (SDE) | 2.99 | 3.98 | 1,000 | ~1M | Good |
| FM – Gaussian Path | Gaussian | 3.02 | 6.9 | 270 | 400K | Excellent |
| FM – OT Path | OT (Straight) | 2.99 | 6.35 | 142 | 400K | Excellent |
| CNF (MLE) | Gaussian | 3.28 | 46 | 150 | ~2M | Poor |

5 rows, 7 columns

## Mathematical Details

### OT Path: Derivation of the Constant Vector Field

For the OT conditional path $x_t = (1-t)x_0 + tx_1$ with $x_0 \sim \mathcal{N}(0, I)$:

$$\mu_t(x_1) = (1-t)x_0 + tx_1, \qquad \sigma_t = 0$$

Differentiating w.r.t. $t$:

$$u_t(x \mid x_1) = \dot{\mu}_t(x_1) = x_1 - x_0$$

The vector field is **constant in time** and simply points from the source sample to the target sample — hence perfectly straight trajectories.

### Log-Density via Change of Variables

After training, exact log-likelihoods are available by integrating the divergence:

$$\log p_1(x_1) = \log p_0(x_0)

```
- \int_0^1 \nabla \cdot v_\theta(x_t, t)\, dt$$
```

This integral is estimated with the **Hutchinson trace estimator** $\nabla \cdot v \approx \mathbb{E}_\epsilon[\epsilon^\top (\nabla_x v)\epsilon]$, requiring only one JVP per time step.

### Proof Sketch of Theorem 2

$$\mathcal{L}*{\text{FM}} - \mathcal{L}*{\text{CFM}} = \mathbb{E}_{t,x}!\left[|u_t(x)|^2 - |u_t(x|x_1)|^2

```
- 2\langle v_\theta, u_t(x) - u_t(x|x_1)\rangle\right]$$
```

The cross term vanishes because $\mathbb{E}_{x_1|x}[u_t(x|x_1)] = u_t(x)$ (the marginalization identity), and both $\|u_t(x)\|^2$ and $\|u_t(x|x_1)\|^2$ are independent of $\theta$. Therefore $\nabla_\theta \mathcal{L}_{\mathrm{FM}}$ =

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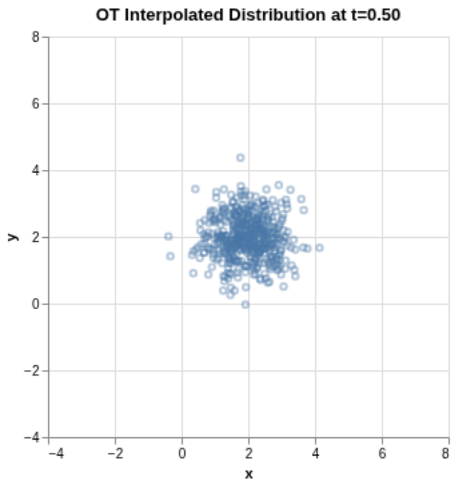
### Interactive: OT Path Distribution Over Time

Use the slider to see how the distribution transitions from $\mathcal{N}(0, I)$ (t=0) to the target cluster (t=1) under the OT path.

Time t ∈ [0,1]




## Sampling Pipeline: Diffusion vs Flow Matching

```
flowchart LR
    subgraph Diffusion_Model
       ND[Noise sample z₁] -->|ScoreNet| SN[Predict score ∇ log p_t]
       SN -->|DDPM Sampler SDE| DS[Refined sample z₀]
    end
    subgraph Flow_Matching
       ND2[Noise sample z₁] -->|FlowNet| FN[Predict vector u_t]
       FN -->|ODE Solver| OS[Refined sample x₀]
    end
```

## Timeline of Key Developments

```
flowchart LR
    A[2015: Score SDE] --> B[2018: CNF]
    B --> C[2020: DDPM]
    C --> D[2021: ScoreFlow]
    D --> E[2022: Rectified Flow / Flow Matching]
    E --> F[2023: Stochastic Interpolants]
```

## Conclusion

### Key Takeaways

1. **Simulation-free training**: CFM regresses against analytic conditional vector fields $u_t(x \mid x_1)$, eliminating the ODE solve in the training loop. Theorem 2 guarantees produces the same gradient as the true FM objective.
2. **OT paths ≈ straight trajectories**: Choosing the OT interpolant $x_t = (1-t)x_0 + tx_1$ gives constant target vectors $u_t = x_1 - x_0$, drastically reducing path curvature needed at inference (142 vs 1 000).
3. **Diffusion is a special case**: Standard DDPM/DDIM corresponds to FM with the Gaussian path; the score is proportional to the CFM target vector field. FM training is general and often more efficient.

4. **Better scaling**: FM trains in 8× fewer iterations than diffusion baselines on ImageNet while achieving equal or better NLL — a direct consequence of the simulation-objective removing per-step solver overhead from training.

## Future Directions

- **Stochastic Interpolants** (Albergo & Vanden-Eijnden, 2023): unify deterministic flows and stochastic diffusion under a single framework with arbitrary endpoints.
- **Mini-batch OT couplings**: approximate the true OT map in minibatches to further straighten trajectories beyond the marginal-OT approximation.
- **Discrete Flow Matching**: extend the framework to discrete token spaces for language and protein-sequence generation.
- **Consistency Models / Shortcut Models**: distill a flow-matched model into a one-step or few-step sampler while preserving sample quality.

## References


- Lipman, Y. et al. (2022). *Flow Matching for Generative Modeling*. ICLR 2023.
- Liu, X. et al. (2022). *Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow*. ICLR 2023.
- Albergo, M. S. & Vanden-Eijnden, E. (2023). *Building Normalizing Flows with Stochastic Interpolants*. ICLR 2023.
- Song, Y. et al. (2021). *Score-Based Generative Modeling through SDEs*. ICLR 2021.
- Ho, J. et al. (2020). *Denoising Diffusion Probabilistic Models*. NeurIPS 2020.